QuickCommerce-InsuranceAI

MATHEMATICAL MODELS DOCUMENT

Section 1 — Disruption Probabilities

P_rain = RainyDays / TotalObservedDays
P_flood = FloodEvents / ObservationPeriod
P_heat = HeatDays / TotalObservedDays
P_pollution = HazardousAQIDays / TotalObservedDays
P_curfew = CurfewEvents / ObservationPeriod
P_strike = StrikeEvents / ObservationPeriod
P_zone = ZoneClosureEvents / ObservationPeriod
P_store = DowntimeHours / TotalOperationalHours
P_server = DowntimeHours / TotalHours
P_traffic = CongestionHours / OperationalHours

Example 1:

RainyDays = 9

TotalObservedDays = 30

P_rain = 9 / 30 = 0.30

Example 2:

FloodEvents = 4

ObservationPeriod = 12 months

P_flood = 4 / 12 = 0.33

Section 2 — Risk Score

RiskScore = Σ (P_i × W_i)

Example:

P_rain = 0.30, W_rain = 0.5

P_flood = 0.33, W_flood = 0.9

RiskScore ≈ 0.30×0.5 + 0.33×0.9 = 0.15 + 0.297 = 0.447

Section 3 — Exposure Score

ES = (Sum of all probabilities) / N

Example:

Total probability sum = 1.6

N = 10

ES = 1.6 / 10 = 0.16

Section 4 — Reliability Score

RS = (ActivityScore + CompletionScore + WorkHistoryScore + FraudScore) / 4

Example:

ActivityScore = 0.80

CompletionScore = 0.95

WorkHistoryScore = 0.50

FraudScore = 0.90

RS = (0.80 + 0.95 + 0.50 + 0.90) / 4 = 0.79

Sub-formulas:

ActivityScore = WeeklyWorkingHours / MaxWorkingHours
CompletionScore = CompletedOrders / AssignedOrders
WorkHistoryScore = min(ExperienceMonths / 24 , 1)
FraudScore = 1 − FraudFlags

Section 5 — Coverage Model

CF = 0.3 + 0.2 × RS + 0.1 × ES
CoverageAmount = WeeklyIncome × CF

Example:

RS = 0.79

ES = 0.16

WeeklyIncome = ₹6000

CF = 0.3 + (0.2×0.79) + (0.1×0.16)

CF ≈ 0.52

CoverageAmount = 6000 × 0.52 = ₹3120

Section 6 — Premium Model

Premium =
WeeklyIncome × CF × BaseRate × (1 + RiskScore) × SubsidyFactor

BaseRate = 0.015 + 0.02 × RiskScore

Example:

RiskScore = 0.48

BaseRate = 0.015 + (0.02 × 0.48)

BaseRate ≈ 0.0246

Premium ≈ ₹32 per week

Section 7 — Payout Model

DailyIncome = WeeklyIncome / 7
PersonalizationFactor = 0.8 + (0.2 × RS) + (0.1 × ES)
PayoutPercent = BaseSeverity × PersonalizationFactor
DurationFactor = EventDurationHours / 24
Payout = DailyIncome × PayoutPercent × DurationFactor

Example:

WeeklyIncome = ₹6000

DailyIncome = 6000 / 7 ≈ ₹857

BaseSeverity (rain) = 0.50

PersonalizationFactor ≈ 1.0

PayoutPercent = 0.50 × 1.0 = 0.50

Duration = 6 hours

DurationFactor = 6 / 24 = 0.25

Payout = 857 × 0.50 × 0.25 ≈ ₹107

Section 8 — Work Loss Model

ExpectedOrders = AvgOrdersPerHour × EventDurationHours

WorkLossRatio =(ExpectedOrders − ActualOrders) / ExpectedOrders

Example:

AvgOrdersPerHour = 3

Duration = 4 hours

ExpectedOrders = 3 × 4 = 12

ActualOrders = 3

WorkLossRatio = (12 − 3) / 12 = 0.75

Section 9 — Claim Condition

WorkLossRatio ≥ 0.5

✓ Claim Approved

❌ Otherwise Rejected

Section 10 — Financial Metrics

TotalPremiumCollected = Σ WeeklyPremiums

TotalPayout = Σ ClaimPayouts

ProfitOrLoss = TotalPremiumCollected − TotalPayout

LossRatio = TotalPayout / TotalPremiumCollected